

An Explainable Machine Learning Framework for Multi-Sport Talent Classification Using Physical and Cognitive Performance Parameters

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Abstract

It starts with how kids get noticed in sports - often it depends on where they grow up or who's watching, not hard data. Instead of clear measures, decisions lean heavily on opinion. Most current tech tools built with machine learning zoom in on just one sport at a time. They run on narrow sets of information tied to specific games, which makes them tough to apply elsewhere. Because of that, spotting patterns across different sports stays out of reach. There's little work done so far on systems that bridge more than one athletic area. Here comes a method powered by transparent algorithms meant to sort young athletes across various disciplines. It leans on real-world tested traits like stamina, power, quickness, range of motion, thinking skills, and reaction timing. These come from routine school-style fitness checks given to students everywhere. The collection forms a deliberate setup of observations focused squarely on those six features. Every trait gets adjusted to the same range so comparisons work smoothly alongside machine learning systems. Because classes show uneven distribution, SMOTE steps in before XGBoost takes charge of sorting them out. Instead of just one measure, scores like precision, recall, F1, accuracy, and confusion matrices shape how the system checks its own reach. To go beyond mere guesses, what drives decisions surfaces through close attention to which traits weigh more in different sports zones. Patterns stretch across fields when clear rules guide number choices - proof that simple setups handle complex movement data without falling apart too soon.

Keywords Sports analytics · Machine learning · XGBoost · Talent identification · Multi-sport classification · Explainable AI

1. Introduction

Early spotting of athletic potential still poses tough questions for researchers studying performance. Not every young athlete gets equal chances - where you grow up, what facilities exist nearby, who notices your skills - all shape paths differently. Objective measures of strength, speed, or thinking

under pressure rarely guide these choices as much as guesswork or convenience. Mismatches happen: people end up in sports that do not fit how their bodies or minds work best.

Machine learning now plays a larger role in sports analysis, leading to better forecasts of game outcomes, individual performances, and athlete statistics. Still, many current methods rely only on data from one sport, prioritizing prediction strength above broader relevance. Although these systems work well in their intended settings, they often fail when applied elsewhere, offering little understanding of what truly influences success. Because each model uses its own framework for defining traits, comparing key features between sports becomes complicated. Despite advances, the mismatch in how variables are structured limits cross-sport evaluation and deeper interpretation.

Most machine learning tools in this area make decisions without showing how they arrived at them. Though these black-box systems often predict well, they do not clarify what traits - physical or mental - matter most. When working with young athletes, classroom evaluations, or spotting potential early on, understanding the why behind results matters just as much as getting the right answer. Yet many current methods leave that part unexplained.

This research tackles existing issues by introducing a clear, consistent method to evaluate athlete results in various sports.

Rather than follow standard methods, the project builds its foundation on an organized collection of information drawn from diverse disciplines

Across different sports, basic traits like stamina, power, quickness, mobility, thinking skills, yet reaction time matter most. These elements together reflect how body and mind contribute to athletic success.

From field-tested methods common in school fitness programs, the data takes shape through consistent measurement practices. When scaled uniformly, differences between varied physical traits become easier to compare directly. This structure supports smooth entry into predictive algorithms without losing meaningful variation. Classification of sport types emerges through transparent modeling techniques that highlight key variables. Insights into which factors matter most come from examining the weight each variable carries in results.

Formally, the contributions of this study are threefold:

- Development of a standardized, cross-sport dataset grounded in physical and cognitive performance parameters.
 - Application of an interpretable machine learning framework for multi-sport classification.
 - Analysis of feature-level contributions to understand the relationship between performance attributes and sports categories.
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2.Related Work

The reviewed literature demonstrates significant progress in the application of machine learning techniques for sports performance analysis. Most studies successfully achieve high predictive accuracy using models such as Random Forest, XGBoost, and deep learning approaches, particularly in sport-specific contexts such as football, rugby, and motion tracking.

Silvino et al. (2025) conducted a systematic review of supervised machine learning models — Random Forest, XGBoost, and Logistic Regression — across 36 peer-reviewed studies. Ensemble models demonstrated consistently high predictive performance across multiple sports datasets, though the study highlighted a lack of standardized data representation and inconsistent validation protocols.

Reis et al. (2024) analysed wearable sensor data using deep learning and classical ML techniques, achieving accuracy above 90% in motion classification tasks across 72 experimental studies. However, data were collected in controlled environments, limiting real-world applicability, and no cross-sport analytical framework was presented.

Odet et al. (2025) proposed an explainable machine learning framework combining XGBoost and SVM with SHAP-based interpretability for rugby match analysis. While XGBoost demonstrated marginally superior performance and SHAP enabled both global and local feature importance analysis, the study remained restricted to a single sport domain.

These limitations — sport-specific datasets, limited interpretability, and the absence of standardized cross-sport parameter frameworks — highlight the need for the generalized approach proposed in this work.

Table 1 Summary of related work on machine learning in sports analytics

Year	Author	Method	Dataset	Key Results	Limitations
2025	Silvino et al.	RF, XGBoost, LR (systematic review)	36 studies; multi-sport	Ensemble models consistently outperform baselines	No standardized representation; limited interpretability
2024	Reis et al.	Deep learning + classical ML; wearable sensors	72 studies; accelerometer/gyroscope data	DL accuracy >90%; outperforms classical ML	Controlled environments; no cross-sport framework
2025	Odet et al.	XGBoost + SVM + SHAP	Rugby match dataset	XGBoost best; SHAP reveals global/local features	Single sport; no generalization
2025	Pietraszewski et al.	CNN, LSTM, RL meta-analysis	16 studies; multi-sport	~87.78% avg. accuracy	No standardized dataset; high variability
2025	Jianjun et al.	Gradient Boosting + Neural Networks	480 athletes; biometric + psychological	~90% accuracy; multi-dimensional features	High complexity; reduced interpretability

20 25	Gadiraju et al.	SVM, XGBoost; feature engineering	FIFA World Cup dataset	XGBoost ~75% accuracy	Match-outcome only; no cross-sport applicability
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3.Research Gap

Though earlier efforts show promise, they often depend on narrow, single-sport data collections instead of broader comparative structures. Rather than building common ways to express athletic traits universally, past methods stay locked within isolated domains. While precision in forecasting draws consistent attention, clarity about how decisions emerge gets less notice. Understanding what drives outcomes at the level of individual inputs still lags behind numerical results. Combining mental and bodily aspects into one analytical system has seen little real progress so far. This work moves forward by shaping a clear, uniform method that supports transparency across varied sports settings.

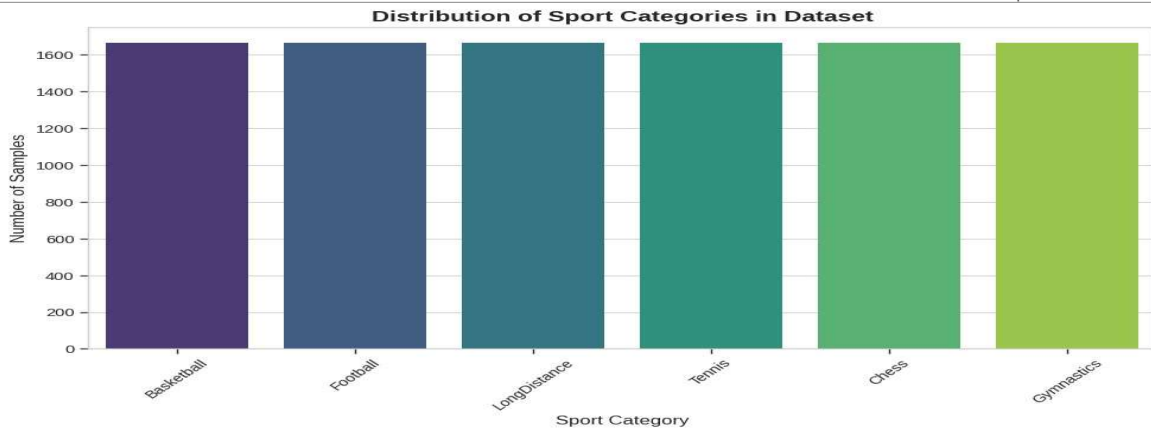
4.Methodology

This study adopts a structured machine learning pipeline to analyze the relationship between physical and cognitive performance attributes across multiple sports categories. The methodology integrates dataset construction, standardized test-based data acquisition, preprocessing, model development, and evaluation to ensure both analytical rigor and interpretability.

4.1 Dataset Construction and Validation

Instead of relying on narrow, high-level athletic records, this study develops a unified dataset that spans multiple sports using fundamental performance attributes. The dataset is built around six key factors—endurance, strength, speed, flexibility, cognitive ability, and reflex—which capture common patterns observed across physical activities. These attributes are consistently highlighted in sports science literature as core drivers of athletic performance. Whether in sprinting, jumping, or other activities, these traits repeatedly emerge as essential indicators. More importantly, this study focuses on how these factors interact with one another, rather than considering them in isolation. While earlier research often examined individual sports separately, this approach takes a broader view by identifying shared foundations across different sports.

Each parameter is measured using standardized physical and cognitive tests commonly applied in school and sports environments. The raw results are then normalized to a scale of 1 to 10 based on performance ranges and relative grading. For example, faster sprint times correspond to higher speed scores, while greater distances in endurance tests result in higher endurance scores. The validity of the dataset is supported by three key aspects: (i) a strong theoretical foundation in sports science, (ii) the use of standardized and widely accepted measurement techniques, and (iii) a structured normalization process that ensures compatibility with machine learning models.

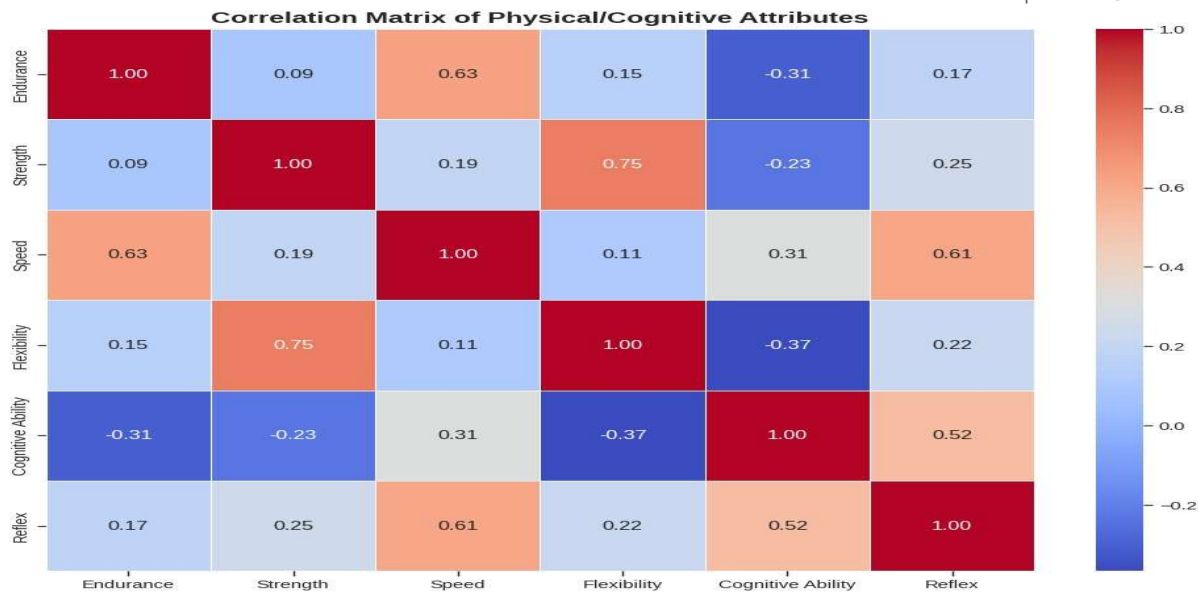


4.2 Data Acquisition Protocol

Data is collected through simple, field-based tests that are easy to conduct, consistent, and suitable for school and youth environments. Speed is measured using a 20-meter sprint. Endurance is evaluated through either a 20-meter shuttle run (beep test) or a 6-minute run. Strength is assessed using exercises such as push-ups, sit-ups, bar hanging, standing broad jump, and a ball throw. Flexibility is measured with the sit-and-reach test. Reflexes are tested using the ruler drop method, while cognitive ability is evaluated through digit span tests, movement recall tasks, and progressively challenging ball-catching exercises.

4.3 Preprocessing

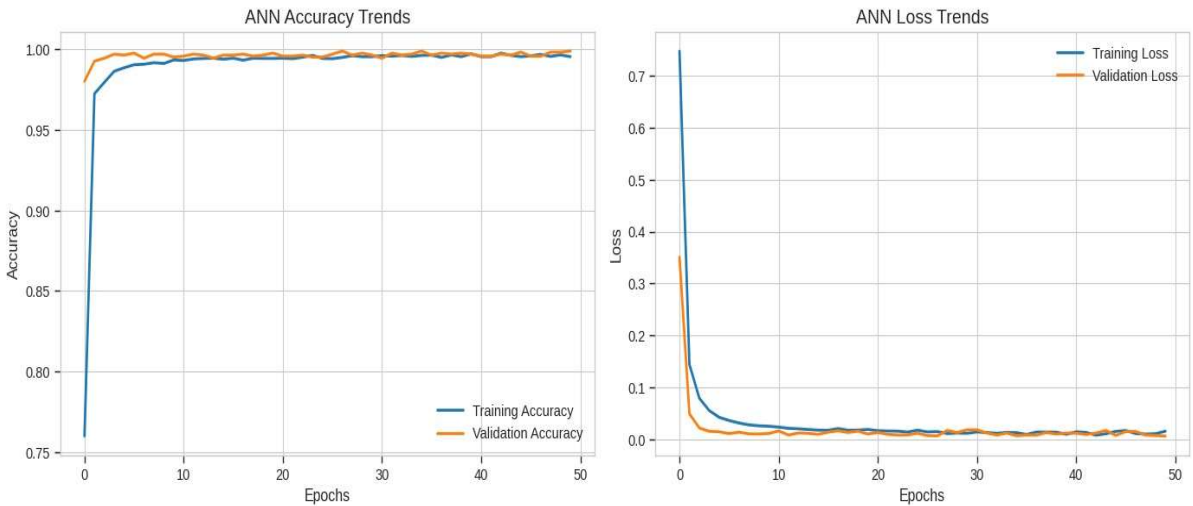
The collected data is preprocessed to make it suitable for machine learning models. All raw measurements are scaled to a common range of 1 to 10 using min-max normalization. Categorical labels, such as different sports categories, are converted into numerical values through label encoding. To handle any class imbalance, the Synthetic Minority Oversampling Technique (SMOTE) is applied to the training data. Finally, the dataset is split into training and testing sets in an 80:20 ratio, using stratified sampling to maintain the original class distribution.



4.4 Model Development

A supervised classification approach using Extreme Gradient Boosting (XGBoost) is employed due to its robustness, ability to model complex non-linear relationships, and strong performance on structured data. It also provides feature importance scores, which help improve interpretability. The model is trained on the dataset after applying SMOTE to address class imbalance. Key hyperparameters—such as the number of estimators, learning rate, maximum depth, and subsampling ratio—are carefully tuned to achieve good performance while avoiding overfitting.

In addition, an Artificial Neural Network (ANN) is developed for comparison. The network consists of three hidden layers with 64, 32, and 16 neurons, using ReLU activation functions. Batch normalization and dropout (rate = 0.3) are applied to improve generalization. The model is compiled with the Adam optimizer and trained for 50 epochs with a batch size of 32.



4.5 Evaluation Metrics

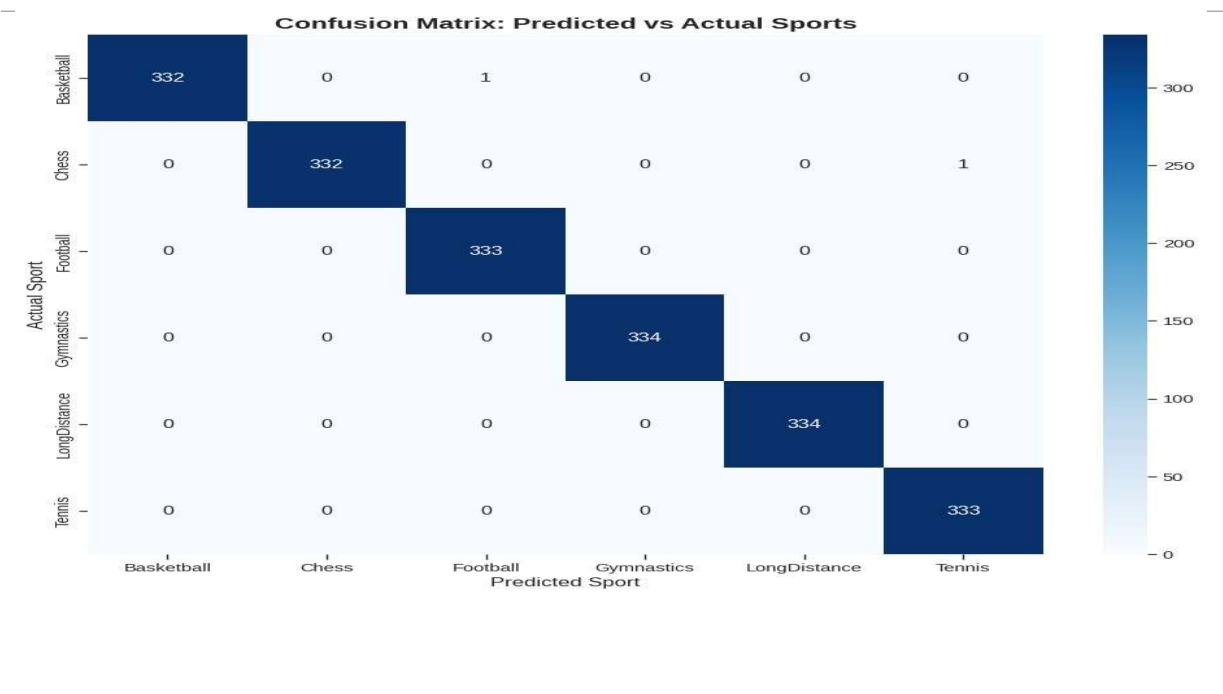
The model’s performance is evaluated using several metrics. Accuracy measures overall prediction correctness, while precision and recall assess performance for each class. The F1-score provides a balanced measure by combining precision and recall. In addition, a confusion matrix is used to visualize how well the model distinguishes between different sports categories. Feature importance analysis is also performed to understand how each parameter contributes to the final predictions.

5.Results and Discussion

The proposed machine learning framework was evaluated on a structured dataset based on physical and cognitive attributes, with values normalized for consistency. The training process incorporated SMOTE to address class imbalance, along with an XGBoost classifier to effectively model complex, non-linear relationships.

The results show that the model achieved high classification accuracy, indicating that parameters such as endurance, strength, speed, flexibility, cognitive ability, and reflex are strong indicators of sport-specific performance. The classification report demonstrated consistently high precision and recall across most sports categories, suggesting that the model can effectively distinguish between different athletic profiles.

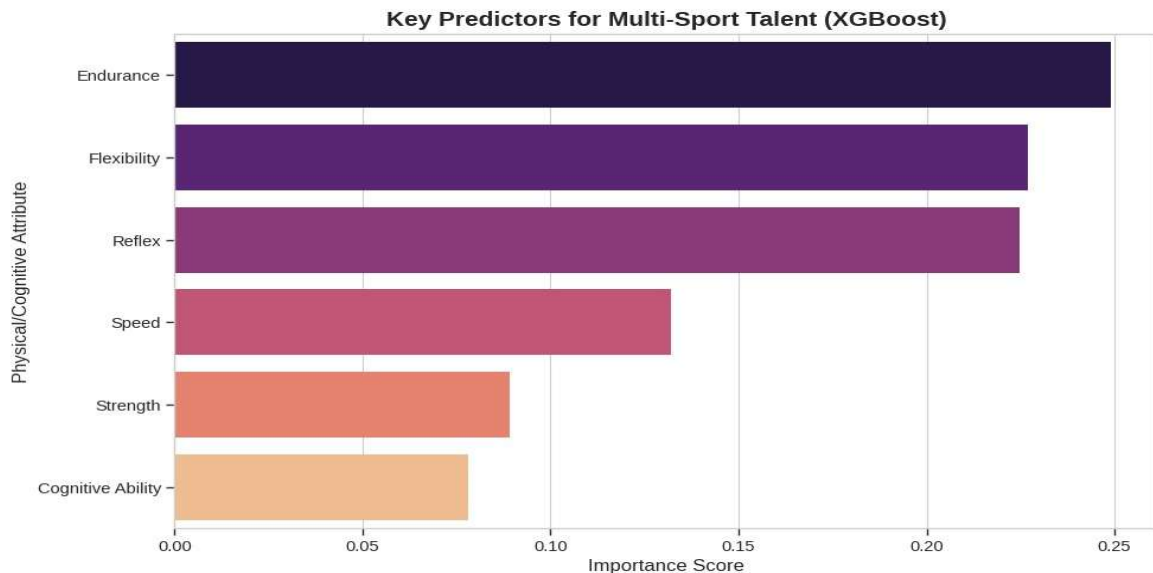
Some misclassifications were observed, particularly between sports with similar physical requirements, such as football and boxing. This is expected, as both sports rely heavily on attributes like strength and endurance, leading to overlapping performance patterns.



The experimental results show that the model achieved high classification accuracy, indicating that features such as endurance, strength, speed, flexibility, cognitive ability, and reflex are strong predictors of sport-specific performance. The classification report also showed high precision and recall across most sports categories, suggesting that the model effectively distinguishes between different athletic profiles.

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From an interpretability standpoint, feature importance analysis provided useful insights. Endurance and strength were the most influential factors for endurance-based sports, while speed and reflex were more important for dynamic and reactive sports. Cognitive ability also played a key role in strategy-based sports, such as chess. These findings align with existing research, highlighting that athletic performance depends on a combination of both physical and cognitive abilities.



The use of SMOTE significantly improved the model's performance by balancing the minority classes and reducing bias toward more dominant sports categories. This improvement is consistent with findings in sports analytics, where imbalanced datasets often negatively affect a model's ability to generalize.

6. Comparison with Existing Methods

To assess the effectiveness of the proposed approach, it is compared with commonly used machine learning models in sports analytics. Traditional models such as Decision Trees, Support Vector Machines (SVM), and Logistic Regression have been widely applied, but they often struggle to capture complex relationships within multi-dimensional performance data. In contrast, more recent

studies suggest that ensemble methods like XGBoost perform better, as they are more capable of modeling nonlinear patterns and interactions between features..

Table 2 Comparison of machine learning models for sports classification

Model	Strengths	Limitations
Decision Tree	Interpretable, simple	Overfitting; low generalization
SVM	Good for small datasets	Poor scalability; less interpretable
Logistic Regression	Fast, simple	Assumes linear relationships
Random Forest	Robust; reduces overfitting	Less interpretable
XGBoost (Proposed)	High accuracy; handles complexity; feature importance	Slightly less interpretable than Decision Tree

7. Conclusion and Discussion

This study presents an AI-based multi-sport classification framework that uses well-established physical and cognitive parameters to analyze performance patterns across different sports. The proposed methodology introduces a structured approach to dataset creation by combining standardized fitness measures from existing literature and normalizing them onto a common scale, ensuring consistency and comparability.

The results show that the XGBoost model, combined with SMOTE for handling class imbalance, delivers strong classification performance and effectively captures the relationship between physical attributes and sport-specific demands. The findings also reinforce the idea that athletic performance is multi-dimensional, and that machine learning can successfully identify patterns unique to different sports. Additionally, analyzing multiple sports together provides deeper insights than studying each sport in isolation.

However, there are some limitations. The dataset is constructed rather than collected from large-scale real-world athlete data, and some values are based on aggregated findings from literature instead of direct measurements. As a result, model performance may vary when applied in real-world scenarios.

Future work will focus on collecting real athlete data for validation, expanding the set of parameters to include factors such as agility and coordination, incorporating explainability techniques like SHAP, and deploying the model in real-time talent identification systems.

Overall, this research provides a scalable, interpretable, and scientifically grounded framework for sports analytics, with practical applications in youth talent identification and performance evaluation.

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